

Ultrasonic Tracking

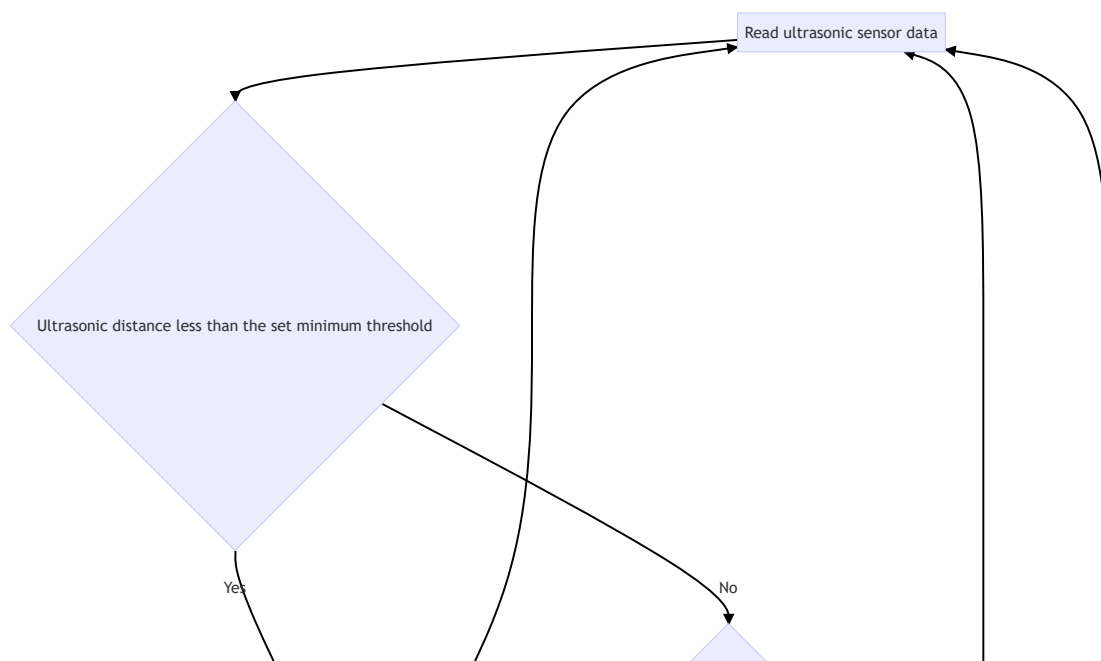
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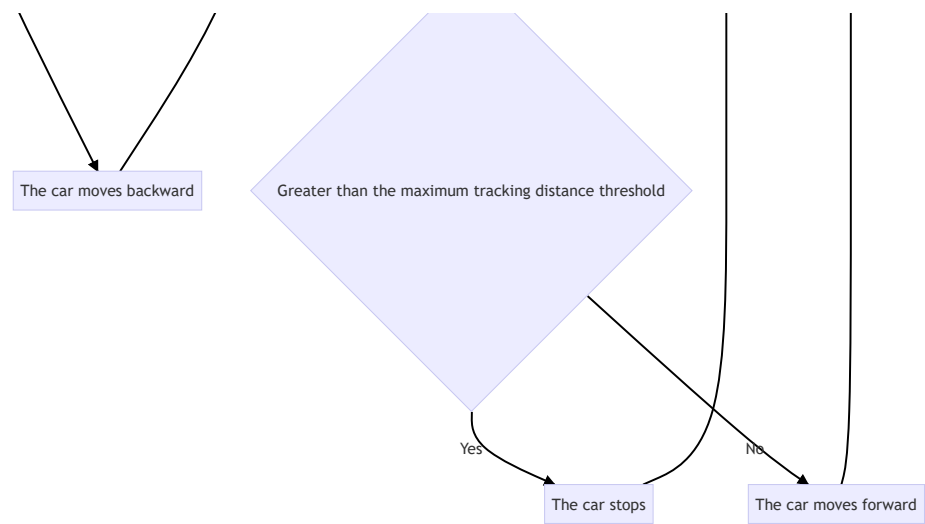
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1. Experiment Purpose

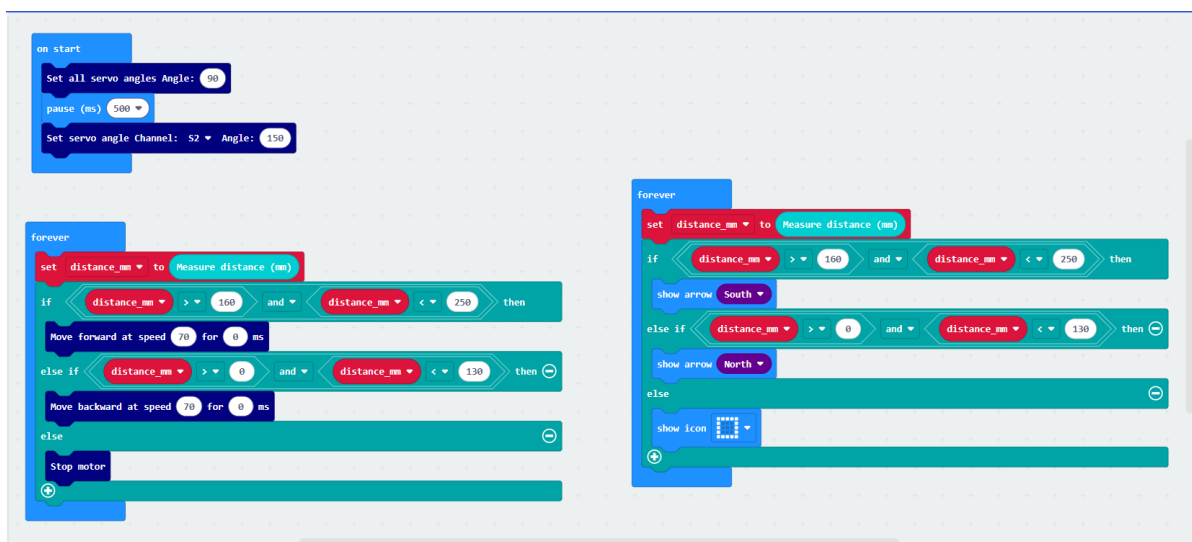
In this section, we will use the ultrasonic sensor together with motor control to achieve the "ultrasonic tracking" effect. The car will automatically adjust its movement state according to the distance to the object in front.

2. Implementation Principle





The program is as follows.



The positions of the blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `17.Ultrasonic_Tracking.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

After flashing the program, the ultrasonic sensor detects the distance to the object ahead. When the object moves away from the car, the car moves forward to track the object; when the object approaches the car, the car moves backward to keep a certain distance from the object.



